

Models of Machine Intelligence Beyond the Brain-Computer Metaphor

The dominant paradigm in AI research and development is fundamentally shaped by the brain-computer metaphor, which regards intelligence as a form of information processing, akin to computational software running on the hardware of the brain. This paper argues that this metaphor, whilst productive in narrow domains, is critically limiting. It reduces the rich, contextual phenomenon of intelligence to abstract computation, constrains the developmental paths of AI by funnelling research towards a single model, and embeds specific Western cultural biases that are presented as universal truths. To move beyond these limitations, this paper proposes alternative frameworks to offer pathways towards developing AI systems that are more ethical, contextual, and socially beneficial.

The brain-computer metaphor, stemming from mid-20th-century cybernetics and sustained by military and corporate funding, faces profound philosophical, scientific, and practical challenges. Philosophical critiques reveal fundamental flaws in the notion of disembodied, computational intelligence. Drawing on Heidegger and Merleau-Ponty, Dreyfus argued that human expertise is not a matter of internalising and processing abstract rules. Instead, it is a form of embodied, situated 'know-how', a skilled engagement with the world that is intuitive and context dependent (Dreyfus, 2000). A master chess player, for instance, perceives the board in terms of threats and opportunities (a 'gestalt'), not as a discrete set of calculable positions. This critique challenges the very foundation of rule-based and statistical AI, highlighting what it misses: the embodied ground of intelligent action.

Additionally, Searle's thought experiment demonstrates that syntactic manipulation (following rules for manipulating symbols) is not sufficient for semantic understanding (grasping meaning) (Cole, n.d.). A person in a room following instructions to manipulate Chinese characters could produce coherent responses without understanding a word of Chinese. Similarly, a computer processing symbols according to its program may simulate intelligence but lacks any genuine understanding or intentionality. This underscores a critical gap between computation and consciousness.

Moreover, Wittgenstein showed that language is not a formal logic system, but a toolkit used within specific forms of life and social practices (Biletzki & Matar, 2021). A word's meaning is its use in a particular context, not an abstract definition. This challenges AI's attempt to capture meaning through formal ontologies and rule-based systems, as language's infinite contextuality and social embeddedness inevitably escape any fixed formalisation.

We may also look to cognitive science for critiques. Modern cognitive science has largely abandoned the computational model of the mind, further undermining the metaphor's scientific basis.

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Lakoff and Johnson argue that our conceptual system and reasoning are grounded in our sensorimotor experiences (Lakoff & Johnson, 2003). We understand abstract concepts (e.g., 'support,' 'distance,' 'affection') through metaphorical extensions of physical experiences. This suggests that cognition is not disembodied abstraction but is fundamentally shaped by the particularities of our bodies.

Moreover, Suchman's work demonstrated that human action is not the execution of a pre-formed plan but is 'situated', improvised in real-time in response to a dynamically changing environment (Suchman, 1987). This contrasts sharply with the AI model of intelligence as the internal processing of a world model followed by plan execution.

Finally, Clark and Chalmers propose that the mind is not confined to the skull but extends into the environment (Clark et al., 1998). A notebook, a smartphone, or even the arrangement of objects can be constitutive parts of a cognitive process. This 'extended mind' view challenges the computer metaphor's focus on internal processing and suggests intelligence is a distributed system involving brain, body, and world.

In addition, these theoretical limitations manifest as persistent, unsolved problems in AI engineering. First, there is the issue of brittleness. State-of-the-art AI systems, including deep learning models, are notoriously brittle (Druce et al., 2021). They can majorly fail when confronted with novel situations or slight alterations to input data (adversarial examples) that a human would effortlessly navigate. This highlights the lack of robust, context-sensitive understanding.

Again, common-sense reasoning gaps persist. Despite vast datasets and computational power, AI systems lack the basic, tacit understanding of the world that humans acquire through embodied interaction (Richardson & Heck, 2023). Knowing that water is wet, that people have interior lives, or that a glass on the edge of a table might fall remains an enormous challenge.

Finally, we see value alignment problems even in advanced AI systems. Encoding human values into machines is difficult precisely because our values are not a set of computable rules (Gabriel, 2020). They are nuanced, contextual, often contradictory, and deeply embedded in our cultural and biological existence. Reducing ethics to optimisation functions risks creating systems that pursue stated goals with literal-minded and catastrophic consequences.

To overcome these limitations, we must look beyond the Western computational tradition to richer, more holistic conceptions of intelligence. First, we may look to non-Western epistemologies. Many cultures understand knowledge and intelligence in relational, ecological, and practice-based ways. For example, the Indigenous Knowledge System shows promise for addressing some of these issues. Navajo philosophy, which centres on *hózhó*, a concept of beauty, balance, and harmonious relations (Diné & Koithan, 2015). Intelligence here is about maintaining balance within a

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network of relationships (human and non-human), not extracting patterns from data (Diné & Koithan, 2015).

Again, we may look to East Asian Philosophies for solutions. Daoist wuwei (effortless action) describes an intelligent, spontaneous responsiveness to a situation, akin to a skilled craftsman's flow state (Hansen, 2025). It is the opposite of top-down, calculative control. Buddhist philosophy emphasises dependent origination, the idea that all phenomena arise in dependence upon a web of causes and conditions (Bhattacharya, 1982). This suggests a model of intelligence as understanding and navigating interdependent systems, not analysing isolated entities.

Additionally, African Philosophy's metaphors of intelligence appear to mitigate some concerns. Ubuntu, often translated as "I am because we are", locates intelligence and humanity within community and mutual relationships (Mungwini et al., 2022). An AI informed by Ubuntu would be evaluated on its capacity to strengthen community bonds and foster reciprocity, not on its individual processing power.

We can also look at the Embodied and Ecological approaches. These Western-adjacent frameworks align with non-Western views in prioritising the role of the body and environment.

For example, there is 4E Cognition. This framework consolidates the scientific shift, viewing cognition as Embodied (shaped by the body), Embedded (dependent on the environment), Extended (leaning on external tools), and Enactive (bringing forth a world of meaning through action) (Alexander & Alexander, 2025). This provides a robust scientific foundation for alternatives to computationalism.

Additionally, Gibson's Affordances Theory has impressive explanatory power. Psychologist J.J. Gibson proposed that we perceive the world directly in terms of possibilities for action ('affordances'), a chair 'affords' sitting, a handle 'affords' grasping (Wells, 2002). This suggests intelligence is about perceiving and acting on these relational possibilities, not building internal representations.

Finally, feminist theory provides crucial critiques of the power dynamics embedded in technology and proposes ethics-centred alternatives. For example, Donna Haraway argues that all knowledge is produced from a particular, situated perspective (Haraway, 1988). The dream of a neutral, objective 'view from nowhere' (which underpins the computational metaphor) is an illusion that obscures its own biases. Instead, she calls for accountable, partial perspectives that recognise their positionality.

Additionally, we may consider the Ethics of Care. Pioneered by Gilligan and Noddings, this framework prioritises responsibilities within relationships over abstract, impartial rules (Maureen Sander-Staudt, n.d.). It values empathy, responsiveness, and nurturing.

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Applying an ethics of care to AI would mean designing systems that are attentive to needs, context, and the specific relationships they are part of.

Building on these alternative frameworks, we can propose new guiding metaphors for machine intelligence.

First, intelligence as Conversation, or Dialogical Exchange. Instead of a processor, imagine an AI as a conversational partner. Drawing from Mikhail Bakhtin, intelligence here is polyphonic, emerging from the dialogue between multiple, independent voices (Gardiner, 2017). Success is measured by the quality of the exchange: the ability to listen, adapt, repair misunderstandings, and contribute meaningfully to a shared understanding. This metaphor prioritises relational skills over processing speed.

Second, Intelligence as Ecological Practice, or as Systemic Relationships. An ecological metaphor views intelligence not as residing in a single agent but as a property of a whole system of relationships. An AI's intelligence would be measured by how it contributes to the health, resilience, and balance of the network it operates within, be it a social community, a business environment, or a natural ecosystem. It shifts the goal from optimisation to integration.

Intelligence as Skilled Material Engagement, or Craft. Michael Polanyi identified 'tacit knowledge', the knowledge that craftspeople and experts embody through practice and cannot easily articulate into rules (Ray, 2009). An AI modelled on craft would learn through apprenticeship and embodied practice with materials and tools, developing an intuitive 'feel' for its domain. It would be a collaborator, not a replacement for human expertise.

Finally, Intelligence as Responsive Attentiveness, or Care. This metaphor, inspired by Haraway and the ethics of care, defines intelligence as the capacity to notice, attend to, and respond appropriately to the needs of others (Maureen Sander-Staudt, n.d.). A caring AI would be evaluated on its ability to enhance wellbeing, dignity, and flourishing within specific contexts. Its primary function would be support, not optimisation or control.

We may now look at the practical applications of these metaphors. Adopting these alternative metaphors and frameworks would radically alter how we design, evaluate, and govern AI.

First, we may reshape the goals of design to aim at relationality rather than computation. The primary design goal would shift from computational efficiency to fostering healthy relationships between users, communities, and environments. This involves participatory design methods that include diverse stakeholders from the outset.

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Second, we ought to evaluate systems by care metrics rather than just accuracy. Beyond technical benchmarks like accuracy or F1 scores, we would need new metrics like Does the system enhance user wellbeing? Does it strengthen community bonds? Does it promote equity and justice? Does it act with reciprocity?

Third, we can incorporate multiple cultural perspectives in development. This requires actively de-centring Western technocratic perspectives and funding research initiatives led by and for Indigenous, Eastern, and Global South communities, treating their knowledge systems as sources of innovation, rather than ethical constraints.

Finally, we may prioritise environmental and social sustainability. Development would be constrained by the imperative of sustainability, rejecting energy-profligate models and applications that exacerbate social inequality or undermine labour rights. AI should be measured by its long-term impact on people and the planet.

In conclusion, the brain-computer metaphor is not a neutral scientific truth but a historical construct that reflects the specific contexts of its birth: wartime cybernetics, Cold War geopolitics, and Western rationalist philosophy. While it has enabled remarkable narrow applications, its limitations are now clear. It offers a reductive, culturally specific, and ultimately impoverished model of intelligence that constrains our technological imagination. By embracing that all models of intelligence are partial and situated, we can forge a new path. The alternative frameworks explored here, from Indigenous relationality to embodied cognition and feminist ethics of care, provide a fertile ground for this expansion. The new metaphors of conversation, ecology, craft, and care are not mere poetry; they are practical guides that can lead us towards developing AI that is more robust, contextual, and humane.

The key recommendation is to consciously cultivate this pluralism in AI research, funding, and regulation. This means actively supporting research programmes based on alternative metaphors, creating evaluation standards that reflect holistic values, and designing governance structures that are open to multiple cultural perspectives. The goal is not to discard computation, but to situate it within a broader, richer understanding of what intelligence is and what it is for: to foster care, balance, and flourishing for all beings in a shared world.

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